

Unit 3 :: Complex Analysis

☞ PHY0400404: Mathematical Physics ☞

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Syllabus

Unit 3: Complex analysis

Functions of complex variables, analyticity and Cauchy-Riemann conditions, examples of analytic functions; singular functions: poles and branch points, order of singularity; integration of functions with complex variable, Cauchy's integral theorem and Cauchy's integral formula, simply and multiply connected regions; Laurent and Taylor's series expansions, residue theorem with application.

Complex trigonometric and hyperbolic identities

$$1. \cosh x = \frac{e^x + e^{-x}}{2}$$

$$2. \sinh x = \frac{e^x - e^{-x}}{2}$$

$$3. \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$4. \sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

Using the above identities, it can be shown that

$$1. \cosh ix = \cos x$$

$$2. \cos ix = \cosh x$$

$$3. \sinh ix = i \sin x$$

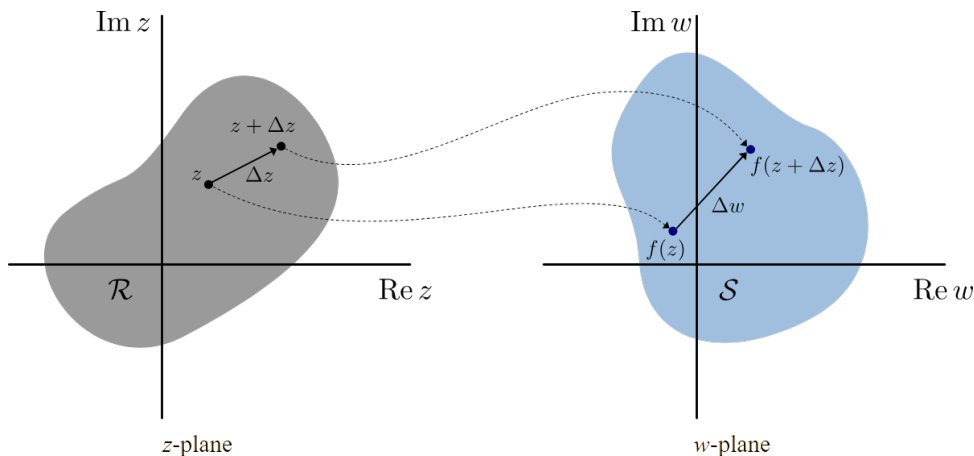
$$4. \sin ix = i \sinh x$$

1 Functions of complex variables

Suppose, to each value that a complex variable z can assume, there corresponds one or more values of a complex variable w . We then say that w is a function of z and write $w = f(z)$.

- The complex variable z is called an *independent variable*
- The complex variable w is called a *dependent variable*.

1.1 Complex derivatives



If a complex function $f(z)$ is single-valued in some region $\mathcal{R}$ of the z plane, the derivative of $f(z)$ is defined as

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

provided that the limit exists independent of the manner in which Δz approaches 0. In such a case, we say that $f(z)$ is *differentiable* at z and $f'(z)$ is known as the *derivative* of the function $f(z)$.

1.2 Complex analytic functions

If the derivative $f'(z)$ exists at all points z of a region $\mathcal{R}$, then $f(z)$ is said to be analytic in $\mathcal{R}$ and is referred to as an *analytic function* in $\mathcal{R}$.

A function $f(z)$ is said to be analytic at a point z_0 if there exists a neighborhood $|z - z_0| < \delta$ at all points of which $f'(z)$ exists.

Note: Analytic functions are sometimes also known as *regular* or *holomorphic* or *monogenic* functions.

1.3 Cauchy-Riemann conditions

Given a complex function which can be written as follows:

$$w = f(z) = u(x, y) + iv(x, y)$$

and is analytic over a region $\mathcal{R}$ on the z -plane, then the 2D functions $u(x, y)$ and $v(x, y)$ satisfy the following Cauchy-Riemann conditions in $\mathcal{R}$:

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}$$

If the partial derivatives of u and v are continuous in $\mathcal{R}$, then the Cauchy-Riemann equations are sufficient conditions that $f(z)$ be analytic in $\mathcal{R}$.

Note: The functions $u(x, y)$ and $v(x, y)$ are known as *conjugate functions*.

Theorem 1

The necessary conditions for a function $f(z) = u(x, y) + iv(x, y)$ to be analytic at all points in a region $\mathcal{R}$ are

a) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$

b) $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

provided that $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ exist.

Theorem 2

The sufficient conditions for a function $f(z) = u(x, y) + iv(x, y)$ to be analytic at all points in a region $\mathcal{R}$ are

- a) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
- b) $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous functions of x and y in the region $\mathcal{R}$.

Theorem 3

Given a function $f(z) = u(x, y) + iv(x, y)$, which is analytic over a region $\mathcal{R}$, its derivative is obtained as

$$f'(z) = \frac{df}{dz} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

1.4 Examples of analyticity of complex functions

1. Determine whether $f(z) = \frac{1}{z}$ is analytic or not.

Solution: Let us consider

$$\begin{aligned} f(z) &= \frac{1}{z} \\ &= \frac{1}{x + iy} \\ &= \left(\frac{1}{x + iy} \right) \times \left(\frac{x - iy}{x - iy} \right) \\ &= \frac{x - iy}{(x + iy)(x - iy)} \\ &= \frac{x - iy}{x^2 + y^2} \\ \Rightarrow f(z) &= \left(\frac{x}{x^2 + y^2} \right) + i \left(\frac{-y}{x^2 + y^2} \right) \\ &= u(x, y) + iv(x, y) \end{aligned}$$

This gives us

$$u(x, y) = \frac{x}{x^2 + y^2} \quad \text{and} \quad v(x, y) = \frac{-y}{x^2 + y^2}$$

We obtain the partial derivatives as

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{(x^2 + y^2) \times 1 - x \times 2x}{(x^2 + y^2)^2} & \frac{\partial u}{\partial y} &= \frac{(x^2 + y^2) \times 0 - x \times 2y}{(x^2 + y^2)^2} \\ &= \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} & &= \frac{-2xy}{(x^2 + y^2)^2} \\ &= \frac{y^2 - x^2}{(x^2 + y^2)^2} \end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial x} &= \frac{(x^2 + y^2) \times 0 - (-y) \times 2x}{(x^2 + y^2)^2} & \frac{\partial v}{\partial y} &= \frac{(x^2 + y^2) \times (-1) - (-y) \times 2y}{(x^2 + y^2)^2} \\
&= \frac{2xy}{(x^2 + y^2)^2} & &= \frac{2y^2 - x^2 - y^2}{(x^2 + y^2)^2} \\
& & &= \frac{y^2 - x^2}{(x^2 + y^2)^2}
\end{aligned}$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous everywhere except at $(0, 0)$.

Therefore the complex function $f(z) = \frac{1}{z}$ is analytic everywhere except at $z = 0$.

2. Show that the function $f(z) = e^x(\cos y + i \sin y)$ is analytic, and then find its derivative.

Solution: Let us consider

$$\begin{aligned}
f(z) &= e^x(\cos y + i \sin y) \\
\Rightarrow f(z) &= (e^x \cos y) + i(e^x \sin y) \\
&= u(x, y) + iv(x, y)
\end{aligned}$$

This gives us

$$u(x, y) = e^x \cos y \quad \text{and} \quad v(x, y) = e^x \sin y$$

We obtain the partial derivatives as

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(e^x \cos y) & \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y}(e^x \cos y) \\
&= e^x \cos y & &= -e^x \sin y \\
\\
\frac{\partial v}{\partial x} &= \frac{\partial}{\partial x}(e^x \sin y) & \frac{\partial v}{\partial y} &= \frac{\partial}{\partial y}(e^x \sin y) \\
&= e^x \sin y & &= e^x \cos y
\end{aligned}$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous everywhere.

Therefore the complex function $f(z) = e^x(\cos y + i \sin y)$ is analytic everywhere.

Finally, the derivative of the given function is obtained as

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ &= e^x \cos y + i e^x \sin y \\ \implies f'(z) &= e^x(\cos y + i \sin y) \end{aligned}$$

3. Test the analyticity of the function $w = \sin z$ and thereby show that $\frac{d}{dz}(\sin z) = \cos z$.

Solution: We have

$$\begin{aligned} w &= \sin z \\ &= \sin(x + iy) \\ &= \sin x \cos iy + \cos x \sin iy \\ &= \sin x(\cosh y) + \cos x(i \sinh y) \\ \implies w &= f(z) = \sin x \cosh y + i \cos x \sinh y \end{aligned}$$

This gives us

$$u(x, y) = \sin x \cosh y \quad \text{and} \quad v(x, y) = \cos x \sinh y$$

The partial derivatives are

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(\sin x \cosh y) & \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y}(\sin x \cosh y) \\ &= \cos x \cosh y & &= \sin x \sinh y \\ \frac{\partial v}{\partial x} &= \frac{\partial}{\partial x}(\cos x \sinh y) & \frac{\partial v}{\partial y} &= \frac{\partial}{\partial y}(\cos x \sinh y) \\ &= -\sin x \sinh y & &= \cos x \cosh y \end{aligned}$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous everywhere.

Therefore the complex function $f(z) = \sin z$ is analytic everywhere.

Now, we evaluate the derivative as follows

$$\begin{aligned} \frac{d}{dz}(\sin z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ &= \cos x \cosh y + i(-\sin x \sinh y) \\ &= \cos x \cosh y - \sin x(i \sinh y) \\ &= \cos x(\cos iy) - \sin x(\sin iy) \\ &= \cos(x + iy) \\ \implies \frac{d}{dz}(\sin z) &= \cos z \end{aligned}$$

4. Find the values of C_1 and C_2 such that the function

$$f(z) = x^2 + C_1 y^2 - 2xy + i(C_2 x^2 - y^2 + 2xy)$$

is analytic. Also show that $f'(z) = 2(1 + i)z$.

Solution: The given function is

$$f(z) = x^2 + C_1 y^2 - 2xy + i(C_2 x^2 - y^2 + 2xy)$$

from which we obtain

$$u(x, y) = x^2 + C_1 y^2 - 2xy \quad \text{and} \quad v(x, y) = C_2 x^2 - y^2 + 2xy$$

For the given function to be analytic, it should satisfy the Cauchy-Riemann conditions, i.e.

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \implies \frac{\partial}{\partial x}(x^2 + C_1 y^2 - 2xy) &= \frac{\partial}{\partial y}(C_2 x^2 - y^2 + 2xy) \\ \implies 2x - 2y &= -2y + 2x \end{aligned}$$

and

$$\begin{aligned} \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \\ \implies \frac{\partial}{\partial y}(x^2 + C_1 y^2 - 2xy) &= -\frac{\partial}{\partial x}(C_2 x^2 - y^2 + 2xy) \\ \implies 2C_1 y - 2x &= -(2C_2 x + 2y) \\ \implies 2C_1 y - 2x &= -2y - 2C_2 x \end{aligned}$$

Comparing the coefficients of y and x above, we obtain

$$\begin{aligned} 2C_1 &= -2 \quad \text{and} \quad -2C_2 = -2 \\ \implies C_1 &= -1 \quad \text{and} \quad C_2 = 1 \end{aligned}$$

Hence the given function is

$$f(z) = x^2 - y^2 - 2xy + i(x^2 - y^2 + 2xy)$$

Now, we have

$$\begin{aligned} \frac{\partial u}{\partial y} &= 2C_1 y - 2x \\ \implies \frac{\partial u}{\partial y} &= -2y - 2x \quad \text{and} \quad \frac{\partial v}{\partial y} = -2y + 2x \end{aligned}$$

Therefore, the derivative is

$$\begin{aligned} f'(z) &= \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \\ \implies f'(z) &= -2y + 2x - i(-2y - 2x) \\ &= -2y + 2x + 2i(x + y) \\ &= 2[(x + iy) + (-y + iy)] \\ &= 2[(1 + i)x + i(1 + i)y] \\ &= 2(1 + i)(x + iy) \\ \implies f'(z) &= 2(1 + i)z \end{aligned}$$

5. Obtain the Cauchy-Riemann conditions in polar form.

Solution: The polar representation of a complex number is

$$z = x + iy = re^{i\theta} \quad (1)$$

Differentiating equation (1) with respect to r and θ give us

$$\frac{\partial z}{\partial r} = e^{i\theta} \quad (2)$$

$$\frac{\partial z}{\partial \theta} = ire^{i\theta} \quad (3)$$

Let us now consider an analytic function given as

$$f(z) = u + iv \quad (4)$$

Differentiating equation (4) with respect to r gives us

$$\begin{aligned} \frac{\partial}{\partial r}[f(z)] &= \frac{\partial}{\partial r}(u + iv) \\ \Rightarrow \frac{df}{dz} \frac{\partial z}{\partial r} &= \frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \\ \Rightarrow f'(z)e^{i\theta} &= \frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \quad [\text{using eqn. (2) on LHS}] \end{aligned}$$

Again, differentiating equation (4) with respect to θ gives us

$$\begin{aligned} \frac{\partial}{\partial \theta}[f(z)] &= \frac{\partial}{\partial \theta}(u + iv) \\ \Rightarrow \frac{df}{dz} \frac{\partial z}{\partial \theta} &= \frac{\partial u}{\partial \theta} + i \frac{\partial v}{\partial \theta} \\ \Rightarrow ir f'(z)e^{i\theta} &= \frac{\partial u}{\partial \theta} + i \frac{\partial v}{\partial \theta} \quad [\text{using eqn. (3) on LHS}] \\ \Rightarrow ir \left[\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right] &= \frac{\partial u}{\partial \theta} + i \frac{\partial v}{\partial \theta} \\ \Rightarrow ir \frac{\partial u}{\partial r} + i^2 r \frac{\partial v}{\partial r} &= \frac{\partial u}{\partial \theta} + i \frac{\partial v}{\partial \theta} \\ \Rightarrow -r \frac{\partial v}{\partial r} + ir \frac{\partial u}{\partial r} &= \frac{\partial u}{\partial \theta} + i \frac{\partial v}{\partial \theta} \quad (5) \end{aligned}$$

Equating the imaginary and real parts on either sides of equation (5), we obtain

$$\begin{aligned} r \frac{\partial u}{\partial r} &= \frac{\partial v}{\partial \theta} & -r \frac{\partial v}{\partial r} &= \frac{\partial u}{\partial \theta} \\ \Rightarrow \frac{\partial u}{\partial r} &= \frac{1}{r} \frac{\partial v}{\partial \theta} & \Rightarrow \frac{\partial u}{\partial \theta} &= -r \frac{\partial v}{\partial r} \end{aligned}$$

Therefore, the Cauchy-Riemann conditions in polar coordinates are

$$\boxed{\begin{aligned} \frac{\partial u}{\partial r} &= \frac{1}{r} \frac{\partial v}{\partial \theta} \\ \frac{\partial u}{\partial \theta} &= -r \frac{\partial v}{\partial r} \end{aligned}}$$

2 Singularities of complex functions

A point at which a complex function $f(z)$ fails to be analytic is called a *singular point* or *singularity* of $f(z)$. Various types of singularities exist, which are discussed below:

2.1 Isolated singularities

A point $z = z_0$ is called an *isolated singularity* or *isolated singular point* of a function $f(z)$ if we can find $\delta > 0$ such that the circle $|z - z_0| = \delta$ encloses no singular point other than z_0 .

- If no such δ can be found, we call z_0 a *non-isolated singularity*.
- If z_0 is not a singular point and we can find $\delta > 0$ such that $|z - z_0| = \delta$ encloses no singular point, then we call z_0 an *ordinary point* of $f(z)$.

Examples

1. $f(z) = \frac{1}{z(z-2)}$ has isolated singularities at $z_0 = 0$ and $z_0 = 2$.
2. $f(z) = e^{1/z}$ has an isolated singularity at $z = 0$.

2.2 Poles

If z_0 is an isolated singularity and we can find a positive integer n such that

$$\lim_{z \rightarrow z_0} (z - z_0)^n f(z) \neq 0$$

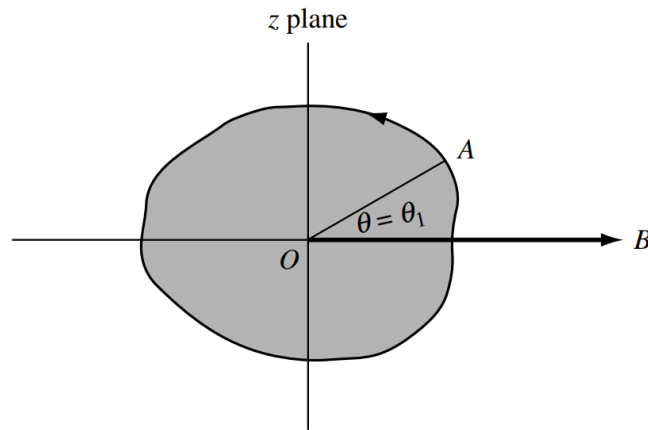
then $z = z_0$ is called a *pole* of order n .

- Special case: If $n = 1$, z_0 is called a *simple pole*.
- If $g(z) = (z - z_0)^n f(z)$, where $f(z_0) \neq 0$ and n is a positive integer, then $z = z_0$ is called a *zero* of order n of $g(z)$. In such a case, z_0 is a pole of order n of the function $\frac{1}{g(z)}$.
 - If $n = 1$, z_0 is called a *simple zero*.

Examples

1. $f(z) = \frac{1}{(z-2)^3}$ has a pole of order 3 at $z_0 = 2$.
2. $f(z) = \frac{3z-2}{(z-1)^2(z+5i)(z-4)}$ has a pole of order 2 at $z_0 = 1$ and simple poles at $z_0 = -5i$ and $z_0 = 4$.

2.3 Branch points



Suppose that we are given the function $w = z^{1/2}$. Further, we allow z to make a complete circuit (counterclockwise) around the origin starting from point A , as shown in the figure above.

We have $z = re^{i\theta}$, and so $w = (re^{i\theta})^{1/2} = \sqrt{r}e^{i\theta/2}$. Then at point A , $\theta = \theta_1$ and $w = \sqrt{r}e^{i\theta_1/2}$. After a complete circuit back to A , $\theta = \theta_1 + 2\pi$ and so

$$\begin{aligned} w &= \sqrt{r}e^{i(\theta_1+2\pi)/2} \\ &= \sqrt{r}e^{i\theta_1/2}e^{i\pi} \\ \implies w &= -\sqrt{r}e^{i\theta_1/2} \end{aligned}$$

Thus, we have not achieved the same value of w with which we started. However, by making a second complete circuit back to A , i.e., $\theta = \theta_1 + 4\pi$, we have

$$\begin{aligned} w &= \sqrt{r}e^{i(\theta_1+4\pi)/2} \\ &= \sqrt{r}e^{i\theta_1/2}e^{i2\pi} \\ \implies w &= \sqrt{r}e^{i\theta_1/2} \end{aligned}$$

and we then do obtain the same value of w with which we started.

We can describe the above by stating that if $0 \leq \theta < 2\pi$, we are on one branch of the multi-valued function $z^{1/2}$, while if $2\pi \leq \theta < 4\pi$, we are on the other branch of the function.

It is clear that each branch of the function is single-valued. In order to keep the function single-valued, we set up an artificial barrier such as OB where B is at infinity which we agree not to cross.

This barrier is called a *branch line* or *branch cut*, and point O is called a *branch point*.

Examples

1. $f(z) = (z - 3)^{1/2}$ has a branch point at $z_0 = 3$.
2. $f(z) = \ln(z^2 + z - 2)$ has branch points where $z^2 + z - 2 = 0$, i.e. at $z_0 = 1$ and $z_0 = -2$.

2.4 Removable singularities

An isolated singular point z_0 is called a *removable singularity* of $f(z)$ if $\lim_{z \rightarrow z_0} f(z)$ exists.

By defining $f(z_0) = \lim_{z \rightarrow z_0} f(z)$, it can then be shown that $f(z)$ is not only continuous at z_0 but is also analytic at z_0 .

Examples

1. $f(z) = \frac{\sin z}{z}$ has a removable singularity at $z_0 = 0$ because $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$, which allows us to define $f(0) = 1$.
2. $f(z) = \frac{e^{z-1} - 1}{z - 1}$ has a removable singularity at $z_0 = 1$ because $\lim_{z \rightarrow 1} \frac{e^{z-1} - 1}{z - 1} = 1$, which allows us to define $f(1) = 1$.

2.5 Essential singularities

An isolated singularity that is not a pole or removable singularity is called an *essential singularity*.

Examples

1. $f(z) = e^{1/(z-2)}$ has an essential singularity at $z_0 = 2$.
2. $f(z) = \sin\left(\frac{1}{z-i}\right)$ has an essential singularity at $z = i$.

3 Complex integration

3.1 Simply and multiply connected regions

3.2 Cauchy's integral theorem

3.3 Cauchy's integral formula

4 Complex series

4.1 Taylor's series expansion of a complex function

4.2 Laurent's series expansion of a complex function

4.2.1 Residue theorem